变分量子蒙特卡洛算法中,基态能量的表达式如下:

$$ E(\theta) = \int E_{loc}(R,\theta) \pi(R,\theta)dR

$$

R 是系统配置,θ 是模型参数 $\pi(R,\theta)$ 是模型在R上的密度: $\pi(R,\theta) = \frac{ \left| \psi(R,\theta) \right|^2}{\int \left| \psi(R,\theta) \right|^2 dR}$ $E_{loc}(R,\theta)$ 是本地能量: $E_{loc}=\frac{\hat{H}\psi(R,\theta) }{\psi(R,\theta)}$

梯度公式推导为:

$$ \frac{dE(\theta)}{d\theta}=\frac{d}{d\theta} \int E_{loc}(R,\theta) \pi(R,\theta)dR

$$

由于被积函数的上下限不包含θ,根据莱布尼兹求导法则, $\frac{d}{d\theta}$可以放入积分符号内部. 且根据推论: $\frac{df(x)}{dx} = f(x)\frac{ln(f(x))}{dx}$,则:

$$ \nabla_{\theta}E = \int \frac{E_{loc}(R,\theta)}{d\theta} \pi(R,\theta) +E_{loc}(R,\theta) \frac{ln(\pi(R,\theta))}{d\theta} dR $$ $$ \nabla_{\theta}E = \int E_{loc}(R,\theta)\pi(R,\theta) \left[\frac{d\ln(\pi(R,\theta))}{d\theta}+\frac{dE_{loc}(R,\theta)}{d\theta} \right] dR $$ 其中$\int{\frac{d\ln(E_{loc}(R,\theta))}{d(\theta)} \pi(R,\theta) dR} = 0$

$$ \therefore \nabla_{\theta}E = \int E_{loc}(R,\theta)\pi(R,\theta) \nabla_{\theta}\ln{\pi(R,\theta)} dR $$

$$\nabla_{\theta}\ln(\pi(R,\theta)) = \frac{d}{d\theta}\left[\ln{\psi(R,\theta)\psi^(R,\theta)}-\ln{\int\left| \psi(R,\theta) \right|^2 dR} \right] $$ $$ 令Z = \int \left| \psi(R,\theta) \right|^2 dR\ \nabla_{\theta}\ln{Z} = \frac{1}{Z} \nabla_{\theta}Z = \frac{1}{Z} \frac{d}{d\theta} \int \left| \psi(R,\theta) \right|^2 dR\ \frac{1}{Z} \nabla_{\theta} \int{\psi(R,\theta) \psi^(R,\theta) dR}=\frac{1}{Z} \int{\nabla_{\theta}\psi \psi^* + \nabla_{\theta}\psi^* \psi dR} $$ 因此整合得到:

$$ \nabla_{\theta}E = \int{E_{loc}\pi \left[\nabla\ln{\psi} + \nabla \ln{\psi^* - \frac{1}{Z}\int{\nabla \psi \psi^* +\nabla \psi^* \psi}} \right]} dR $$

令$\mathcal{O}{\psi} = \nabla{\theta} \ln{\psi(R,\theta)}$,令$\mathcal{O}{\psi^} = \nabla*{\theta} \ln{\psi^(R,\theta)}$，所以$\mathcal{O}{\psi} + \mathcal{O}{\psi^} =\frac{\nabla \psi \psi^* + \nabla \psi^* \psi}{Z}$

$$ \therefore \nabla_{\theta}E = \int{E_{loc}\pi \left[（\mathcal{O}{\psi} + \mathcal{O}{\psi^}） - \frac{1}{Z}\int{\nabla \psi \psi^ +\nabla \psi^* \psi}dR^{\prime} \right]} dR $$ 总

之最后的梯度公式为: $$ \boxed{ \nabla_{\theta} E = \mathbb{E}{\pi}\left[E{\text{loc}} \left(\mathcal{O}{\psi} + \mathcal{O}{\psi^} \right) \right] - \mathbb{E}{\pi}\left[E*{\text{loc}} \right] \mathbb{E}{\pi}\left[\mathcal{O}{\psi} + \mathcal{O}{\psi^*} \right] } $$

在代码里,将变形为: $$ \boxed{ \nabla{\theta}E = \mathbb{E}{\pi}\left[\left(E{\text{loc}}-\mathbb{E}{\pi}\left[E{\text{loc}} \right] \right) \left(\mathcal{O}{\psi} + \mathcal{O}{\psi^*} \right) \right] } $$